

Sarvesh Gharat

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Incoming Visiting Researcher at MBZUAI (2026) — Multi-Agent Systems for Scientific Discovery

EDUCATION

IIT Bombay

PhD in Artificial Intelligence and Data Science

Mumbai, In

July 2022 – May 2027

Vishwakarma Institute of Information Technology

BTech in Electronics and Telecommunication

Pune, In

Aug. 2018 – May 2022

RESEARCH INTERESTS

My research focuses on online learning, particularly multi-armed bandits and Markov decision processes (MDPs) for sequential decision-making. I'm also interested in multi-agent systems and currently work on inference-time algorithms along with the bandit theory.

EXPERIENCE

• Summer Research Intern

May 2025 – July 2025

Adobe Research

Host: Soumyabrata Pal, Ramasuri Narayanam

- Improved LLM reasoning under fixed compute constraints by developing a novel method to inject auxiliary information during sequential test-time scaling.
- Designed and implemented an efficient, on-the-fly example selection strategy using embedding similarity, leading to more relevant in-context examples during inference.
- Focused on designing inference-time techniques that enhance model responses without changing any underlying parameters.

Patent Filed: Gharat, S., Pal, S., Narayanam. R., Guided Sequential Test Time Scaling using additional hints from exemplars

• Student Researcher

Oct 2024 – March 2025

Google DeepMind

Host: Aparna Taneja, Milind Tambe

- Quantified the impact of AI-scheduled interventions on beneficiary engagement and health knowledge in a large-scale maternal health program, analyzing data from thousands of participants.
- Demonstrated statistically significant improvements in selected maternal health knowledge and postnatal care behaviors via a rigorous comparative analysis of AI-targeted interventions versus matched control groups.
- Improved reliability of survey-based insights by addressing noise and response variability, enabling statistically significant impact detection.

• Visiting Researcher

Aug 2021 – June 2022

Tata Institute of Fundamental Research

Host: Prof. Shriramesh Prabhu

- Designed optimization methods to estimate the refractive index of optically thin materials using Terahertz Time Domain Spectroscopy.
- Implemented and benchmarked algorithms including Dual Annealing, SHGO, and Newton-Raphson to improve convergence stability.
- Built a user-friendly Python library to support experimental analysis and make the refractive index estimation pipeline accessible for both researchers and educators.

PAPERS UNDER REVIEW

- Gharat, S.,** Komiyama, J., SGHA: Evidence-Grounded Research Problem Discovery with Local Language Models
- Gharat, S.,** The Argmax Gap in Human Chess Move Prediction
- Gharat, S.,** Chavan, U., Karamchandani, N. and Nair, J., Which Model to Train? A Budgeted Bandit Approach to Adaptive Fine-Tuning

- **Gharat, S.**, Karamchandani, N., and Nair, J., Cost-Aware Best Arm Identification via Dueling Feedback with Applications to Large Language Models.
- Kumar, A., **Gharat, S.**, Beyond One-Shot Decisions: Progressive Morphological Reasoning in Galaxy Classification

ACCEPTED PUBLICATIONS

- Singh, V., Ghosal, S., **Gharat, S.**, Pal, S., Narayanam. R., Manocha. D, ThinkRetrieve: Retrieval-Augmented Reasoning Traces for Test-Time Scaling, to appear in Findings of the Association for Computational Linguistics: EMNLP 2026
- **Gharat, S.**, Karamchandani, N. and Nair, J., 2026, May. Cost-Aware Best Arm Identification via Dueling Feedback with Applications to Large Language Models. In Proc. of the 25th International Conference on Autonomous Agents and Multiagent Systems (pp. 3600-3602).
- Dasgupta, A.*, **Gharat, S.***, Madhiwalla, N., Hegde, A., Tambe, M., Taneja, A., 2025, October. AI-Targeted Calls Drive Measurable Improvements in Maternal Health, In Proc. of European Conference on Artificial Intelligence (pp. 5200-5207).
- **Gharat, S.**, Yadav, A., Karamchandani, N. and Nair, J., 2025, April. Representative arm identification: A fixed confidence approach to identify cluster representatives. In ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 1-5). IEEE.
- Suresh, R., Bishnoi, H., Kuklin, A.V., Parikh, A., Molokeev, M., Harinarayanan, R., **Gharat, S.** and Hiba, P., 2024. Revolutionizing physics: a comprehensive survey of machine learning applications. *Frontiers in Physics*, 12, p.1322162.
- Bhatta, G., **Gharat, S.**, Borthakur, A. and Kumar, A., 2024. Gamma-ray blazar classification using machine learning with advanced weight initialization and self-supervised learning techniques. *Monthly Notices of the Royal Astronomical Society*, 528(1), pp.976-986.
- Mistry, P., Prasad, A., Maity, M., Pathak, K., **Gharat, S.**, Lekkas, G., Bhattarai, S., Kumar, D., Lissauer, J.J., Twicken, J.D. and Soubkiou, A., 2024. VaTEST III: Validation of eight potential super-earths from TESS data. *Publications of the Astronomical Society of Australia*, 41, p.e030.
- **Gharat, S.**, Borthakur, A. and Bhatta, G., 2024. Estimation of redshift and associated uncertainty of Fermi/LAT extragalactic sources with Deep Learning. *Monthly Notices of the Royal Astronomical Society*, 527(3), pp.6198-6210.
- **Gharat, S.**, Bose, B., Borthakur, A. and Mazumder, R., 2023. An image processing approach to identify solar plages observed at 393.37 nm by the Kodaikanal solar observatory. *RAS Techniques and Instruments*, 2(1), pp.393-397.
- Mistry, P., Pathak, K., Prasad, A., Lekkas, G., Bhattarai, S., **Gharat, S.**, Maity, M., Kumar, D., Collins, K.A., Schwarz, R.P. and Mann, C.R., 2023. VaTEST. II. Statistical Validation of 11 TESS-detected Exoplanets Orbiting K-type Stars. *The Astronomical Journal*, 166(1), p.9.
- Pininti, V.R., Bhatta, G., Paul, S., Kumar, A., Rajgor, A., Barnwal, R., and **Gharat, S.**, 2023. Exploring short-term optical variability of blazars using TESS. *Monthly Notices of the Royal Astronomical Society*, 518(1), pp.1459-1471.
- **Gharat, S.** and Dandawate, Y., 2022. Galaxy classification: a deep learning approach for classifying Sloan Digital Sky Survey images. *Monthly Notices of the Royal Astronomical Society*, 511(4), pp.5120-5124.
- Kulkarni, J.S., Cengiz, K. and **Gharat, S.**, 2021, March. Compact C-slot microstrip-fed planar antenna for wireless devices. In 2021 International Conference on Emerging Smart Computing and Informatics (ESCI) (pp. 651-654). IEEE.

WORKSHOP PAPERS

- **Gharat, S.**, Komiya, J., 2026. PRISM: Problem Discovery via Structural Motifs in Knowledge Graphs. ICML AI for Science Workshop 2026.
- **Gharat, S.**, , Borthakur, A. and Bhatta, G., 2023. Gamma Ray AGNs: Estimating Redshifts and Blazar Classification using Neural Networks with smart initialization and self-supervised learning. NeurIPS Machine Learning and the Physical Sciences Workshop 2023.

INVITED TALKS & PRESENTATIONS

- **Invited Talks**

- Which Model to Train? A Budgeted Bandit Approach to Adaptive Fine-Tuning
— Cohere for AI Reinforcement Learning Group (2026)
- *Cost-Aware Best Arm Identification via Dueling Feedback with Applications to Large Language Models*
— Cohere for AI Research Connections (2025)
- *Representative arm identification: A fixed confidence approach to identify cluster representatives*
— Cohere for AI Reinforcement Learning Group (2025)
- *Estimating Redshifts of AGNs using Neural Networks*
— IAU-IAA Astroinformatics Seminar (2024)

- **Poster Presentation:**

- ACM India ARCS 2026
- Google Research Week 2024
- IndoML Graduate Forum 2024
- ACM Pingala Interactions in Computing 2024

ACADEMIC SERVICE & HIGHLIGHTS

- **Reviewer / Program Committee**

- IJCAI AI for Social Good (2025, 2026) - Awarded Bronze Tier reviewer distinction in 2026
- KDD AI for Sciences (Cycle1: 2027 Cycle2: 2026)
- AAMAS 2025
- IEEE Transactions on Information Theory

- **Research Grant**

- Awarded *Thinking Lab Research Grant* for the project on "An Online Learning approach for Adaptive Finetuning"
- Selected for the *Cohere Labs Catalyst Grant Program* for a proposal on online learning
- Inaugural cohort grantee of the *Adaption Labs Research Grant*, for research on continual learning, online learning, and adaptive AI systems
- State Bank of India (SBI) funded project (2024–2026) on *Dynamic LLM Optimization and Fine-Tuning via Dueling Bandits*, exploring adaptive model selection under resource constraints

- **Finalist:**

- * Qualcomm Innovation Fellowship 2026-2027
- * Bloomberg Data Science Ph.D. Fellowship program

- **Selected Programs**

- AAMAS Doctoral Consortium 2026
- ACM India ARCS 2026

- Google Research Week (2024, 2025)
- IndoML Graduate Forum 2024
- ACM CODS-COMAD PhD Clinic 2024
- ACM Pingala Interactions in Computing 2024
- **Kaggle Expert**
 - LLM Prompt Recovery Challenge: Silver Medal (Rank 79/2175)
 - Santa 2025 Challenge: Bronze Medal (Rank 228/3357)

TEACHING EXPERIENCE

- **CS 747:** Foundations of Intelligent and Learning Agents (**Best TA Award**)
- **CS 663:** Digital Image Processing
- **EE 325:** Probability and Random Processes
- **DS 203:** Programming for Data Science

MENTORSHIP EXPERIENCE

- Algoverse AI Research Program 2026
- Krittika Astronomy Summer Project, IIT Bombay 2026
- Summer of Sciences, IIT Bombay 2023

TECHNICAL SKILLS

- **Programming:** Python, LaTeX, Apps Script
- **Libraries:** PyTorch, TensorFlow, NumPy, pandas, vLLM, Transformers
- **Tools:** Jupyter Notebook, Git, Docker, Google Cloud Platform, VS Code